TS3000 series Robot Controller

INSTRUCTION MANUAL

I/O CABLE SPECIFICATIONS

Rev. 3
April 1, 2020

Notice

- Make sure that this instruction manual is delivered to the final user of SHIBAURA MACHINE's industrial robot.
- Before installing and operating the industrial robot, read through and completely understand this manual.
- After reading through this manual, keep it nearby for future reference.

Shibaura Machine

COPYRIGHT(C) 2020 BY SHIBAURA MACHINE CO.,LTD.
ALL RIGHTS RESERVED.

No part of this document may be reproduced in any form without obtaining prior written permission from SHIBAURA MACHINE CO.,LTD.

The information contained in this manual is subject to change without prior notice to effect improvements.

PREFACE

This manual gives information about the I/O cables for the robot controllers TS3000 and TS3100.
The I/O cables are optional.
See the instruction manual "INTERFACE" for how to use the cables.

CAUTION

This manual does not describe details on robot connection.

For the details on robot connection, see the instruction manuals "INSTALLATION AND TRANSPORTATION" and "INTERFACE."

CAUTIONS ON SAFETY

This manual contains the important information on the robot and controller to prevent injury to the operators and persons nearby, to prevent damage to assets and to assure correct use.

Make sure that the following details (indications and symbols) are well understood before reading this manual. Always observe the information that is noted.

[Explanation of indications]

Indication	Meaning of indication
DANGER	This means that "incorrect handling will lead to fatalities or serious injuries."
CAUTION	This means that "incorrect handling may lead to personal injuries *1) or physical damage *2)."

- Injuries refer to injuries, burns and electric shocks, etc., which do not require hospitalization or long-term medical treatment.
- Physical damage refers to damages due to destruction of assets or resources.

[Explanation of symbols]

Symbol	Meaning of symbol
	This means that the action is prohibited (must not be done). Details of the actions actually prohibited are indicated with pictures or words in or near the symbol.
	This means that the action is mandatory (must be done). Details of the actions that must be done are indicated with pictures or words in or near the symbol.
	This means danger and caution. The details of the actual danger and caution are indicated with pictures or words in or near the symbol.
	This means caution. The caution details are indicated with pictures or words in or near the symbols.

CAUTION

- Be sure to read through the safety instruction manual provided separately before starting actual work to assure the safe work covering from the robot installation to operation.

[Maintenance and inspection]

Be sure to observe the following items to use the robot safely.

DANGER	
 Mandatory	• Be sure to turn off the main power switch of the controller before starting inspection or maintenance.

CAUTION	
 Disassembly prohibited	The user must NEVER replace or modify parts other than those described in the instruction manual. Otherwise, the performance may deteriorate or faults or accidents will be caused.
 Mandatory	• Always use the SHIBAURA MACHINE's designated spare parts when replacing the parts. • Maintenance and inspection should be performed regularly. Otherwise, the system may malfunction or accidents will be caused.

CAUTION

- To maintain and inspect the robot safely, thoroughly read the instruction manual "MAINTENANCE" as a separate volume before starting the maintenance and inspection.

Table of Contents

1. TS3000/TS3100 SYSTEM CABLE 7
1.1 Specifications 7
1.2 Outline drawing 7
1.3 Wiring table 8
2. TS3000/TS3100 INPUT CABLE 9
2.1 Specifications 9
2.2 Outline drawing 9
2.3 Wiring table 10
3. TS3000/TS3100 OUTPUT CABLE 11
3.1 Specifications 11
3.2 Outline drawing 11
3.3 Wiring table 12

1. TS3000/TS3100 SYSTEM CABLE

1.1 Specifications

Type: 338M07□□

Cable length: L = 6 m (standard)

* See the instruction manual "INTERFACE" (TS3000/TS3100: ST80729) for details on the input signals and connection methods.

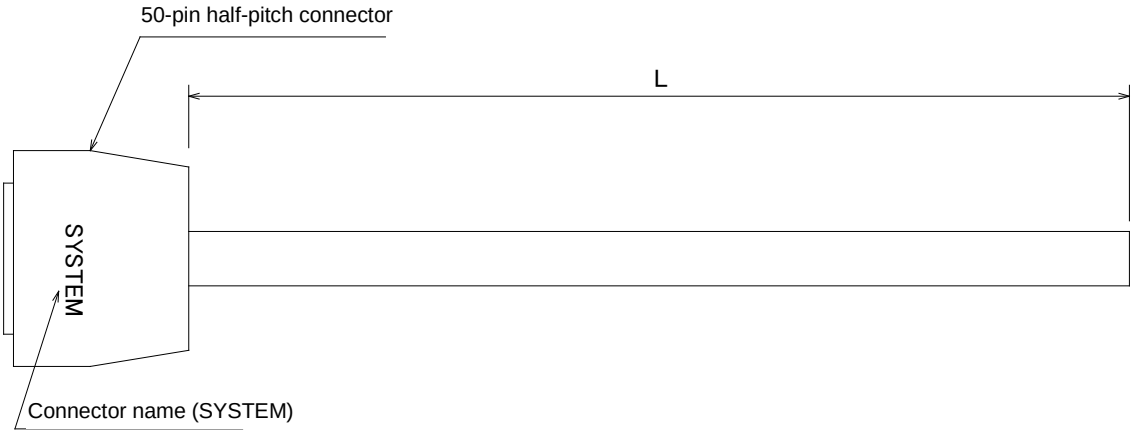
Cable specifications

Item	Specifications
Nominal cross-sectional area of conductor	0.2mm²
Conductor structure	7/0.18
Number of cores	25C
Insulator material	Heat-resistant vinyl chloride
Shield material	Tin-plated soft copper wire
Sheath material	Heat-resistant, oilproof vinyl chloride
Bulk resistance (20°C)	121Ω/km or less
Operating temperature	-20°C to 75°C *1
Rated voltage	50 VAC, 60 VDC

*1: Heat-resistant grade 80°C

*2: The TS3000/TS3100 system cable must not be used with a moving part or sliding part which bends repeatedly.

1.2 Outline drawing



1.3 Wiring table

Signal name	Connector pin	Wire color
Reserved	1	Black/-
Reserved	2	Black/White
Reserved	3	Red/-
ALM_RST/DIN254	4	Red/White
STROBE/DIN249	5	Green/-
PRG_RST/DIN250	6	Green/White
STEP_RST/DIN251	7	Yellow/-
CYC_RST/DIN252	8	Yellow/White
DO_RST/DIN253	9	Brown/-
RUN/DIN255	10	Brown/White
EX_SVON/DIN256	11	Blue/-
STOP/DIN257	12	Blue/White
CYCLE/DIN258	13	Gray/-
BREAK/DIN260	14	Gray/White
LOW_SPD/DIN259	15	Orange/-
SVOFF/DIN261	16	Orange/White
P24G (P24V)	17	Purple/-
P24G (P24V)	18	Purple/White
Reserved	19	Light Green/-
Reserved	20	Light Green/White
Reserved	21	Pink/-
Reserved	22	Pink/White
Reserved	23	Light Blue/-
Reserved	24	Light Blue/White
Reserved	25	White/-
Reserved	26	White/Black
SV_RDY/DOUT250	27	Green/Black
BT_ALM/DOUT261	28	Green/Red
ACK/DOUT251	29	Yellow/Black
TEACH/DOUT252	30	Yellow/Red
EXTSIG/DOUT254	31	Brown/Black
SYS_RDY/DOUT256	32	Brown/Red
AUTORUN/DOUT257	33	Blue/Black
ALARM/DOUT262	34	Blue/Red
CYC_END/DOUT258	35	Gray/Black
LOW_ST/DOUT259	36	Gray/Red
P24V (P24G)	37	Orange/Black
P24V (P24G)	38	Orange/Red
Reserved	39	Purple/Black
Reserved	40	Purple/Red
SVST_A	41	Light Green/Black
SVST_B	42	Light Green/Red
EMSST_A	43	Pink/Black
EMSST_B	44	Pink/Red
P24V	45	Light Blue/Black
P24V	46	Light Blue/Red
P24V	47	Red/Black
P24G	48	Red/Green
P24G	49	Black/Green
P24G	50	Black/Yellow
FG	Case	Shield

2. TS3000/TS3100 INPUT CABLE

2.1 Specifications

Type: M274338

Cable length: L = 6 m (standard)

* See the instruction manual "INTERFACE" (TS3000/TS3100: ST80729) for details on the output signals and connection methods.

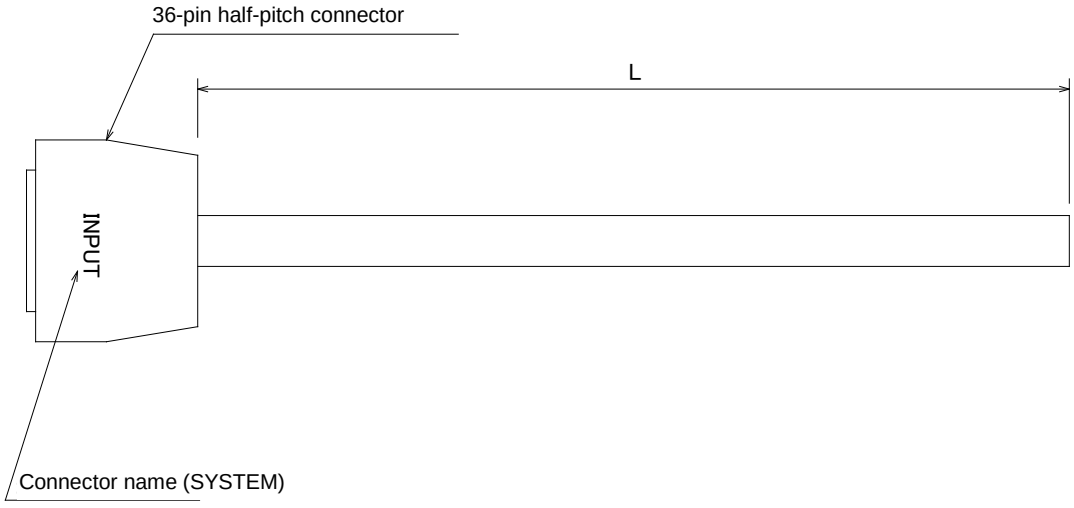
Cable specifications

Item	Specifications
Nominal cross-sectional area of conductor	0.38mm²
Conductor structure	7/0.127 AWG28
Number of cores	25C
Insulator material	Polypropylene
Shield material	Tin-plated soft copper wire
Sheath material	Heat-resistant PVC
Bulk resistance (20°C)	226Ω/km or less
Operating temperature	-0°C to 45°C *1
Rated voltage	30 VAC, 42 VDC

*1: Heat-resistant grade 80°C

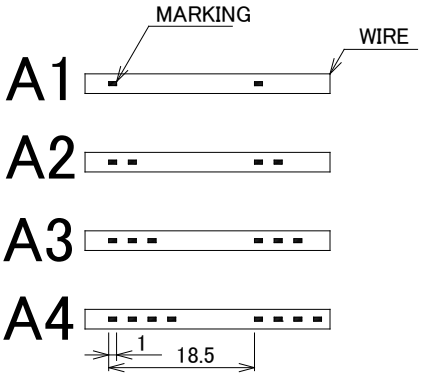
*2: The TS3000/TS3100 system cable must not be used with a moving part or sliding part which bends repeatedly.

2.2 Outline drawing



2.3 Wiring table

Signal name	Connector pin	Wire color/Marking color/dot
DIN1	1	Orange/Red/A1
DIN2	19	Orange/Black/A1
DIN3	2	Gray/Red/A1
DIN4	20	Gray/Black/A1
DIN5	3	White/Red/A1
DIN6	21	White/Black/A1
DIN7	4	Yellow/Red/A1
DIN8	22	Yellow/Black/A1
DIN9	5	Pink/Red/A1
DIN10	23	Pink/Black/A1
DIN11	6	Orange/Red/A2
DIN12	24	Orange/Black/A2
DIN13	7	Gray/Red/A2
DIN14	25	Gray/Black/A2
DIN15	8	White/Red/A2
DIN16	26	White/Black/A2
DIN17	9	Yellow/Red/A2
DIN18	27	Yellow/Black/A2
DIN19	10	Pink/Red/A2
DIN20	28	Pink/Black/A2
DIN21	11	Orange/Red/A3
DIN22	29	Orange/Black/A3
DIN23	12	Gray/Red/A3
DIN24	30	Gray/Black/A3
DIN25	13	White/Red/A3
DIN26	31	White/Black/A3
DIN27	14	Yellow/Red/A3
DIN28	32	Yellow/Black/A3
DIN29	15	Pink/Red/A3
DIN30	33	Pink/Black/A3
DIN31	16	Orange/Red/A4
DIN32	34	Orange/Black/A4
*P24G (P24V)	17	Gray/Red/A4
*P24G (P24V)	35	Gray/Black/A4
*P24G (P24V)	18	White/Red/A4
*P24G (P24V)	36	White/Black/A4
FG	Case	Shield



*Power lines change according to the OUTPUT type. ( ) are for type-P.

3. TS3000/TS3100 OUTPUT CABLE

3.1 Specifications

Type: M274339

Cable length: L = 6 m (standard)

* See the instruction manual "INTERFACE" (TS3000/TS3100: ST80729) for details on the input and output signals and connection methods.

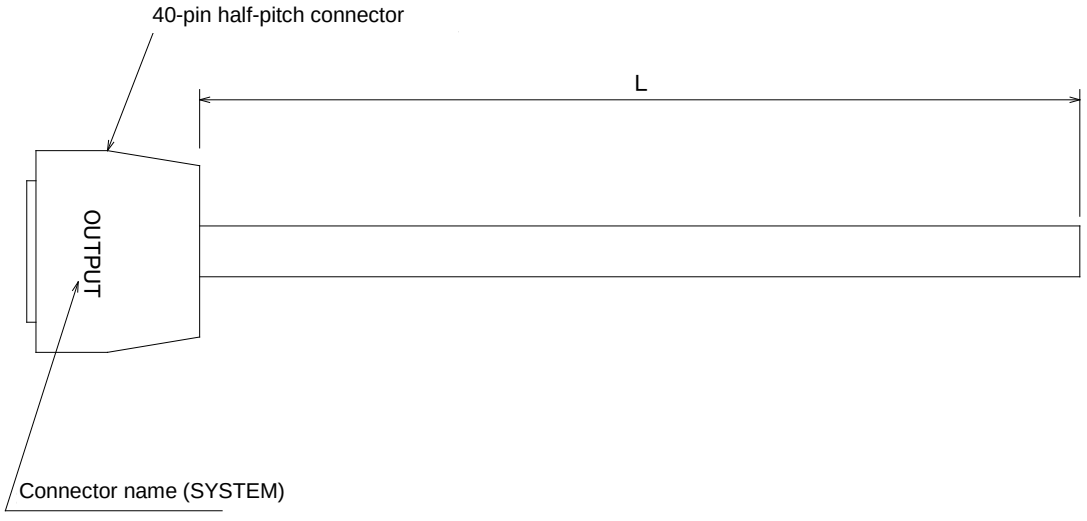
Cable specifications

Item	Specifications
Nominal cross-sectional area of conductor	0.38mm²
Conductor structure	7/0.127 AWG28
Number of cores	25C
Insulator material	Polypropylene
Shield material	Tin-plated soft copper wire
Sheath material	Heat-resistant PVC
Bulk resistance (20°C)	226Ω/km or less
Operating temperature	-0°C to 45°C *1
Rated voltage	30 VAC, 42 VDC

*1: Heat-resistant grade 80°C

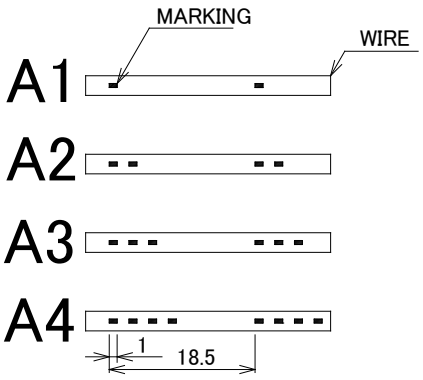
*2: The TS3000/TS3100 system cable must not be used with a moving part or sliding part which bends repeatedly.

3.2 Outline drawing



3.3 Wiring table

Signal name	Connector pin	Wire color/Marking color/dot
DOUT1	1	Orange/Red/A1
DOUT2	21	Orange/Black/A1
DOUT3	2	Gray/Red/A1
DOUT4	22	Gray/Black/A1
DOUT5	3	White/Red/A1
DOUT6	23	White/Black/A1
DOUT7	4	Yellow/Red/A1
DOUT8	24	Yellow/Black/A1
DOUT9	5	Pink/Red/A1
DOUT10	25	Pink/Black/A1
DOUT11	6	Orange/Red/A2
DOUT12	26	Orange/Black/A2
DOUT13	7	Gray/Red/A2
DOUT14	27	Gray/Black/A2
DOUT15	8	White/Red/A2
DOUT16	28	White/Black/A2
DOUT17	9	Yellow/Red/A2
DOUT18	29	Yellow/Black/A2
DOUT19	10	Pink/Red/A2
DOUT20	30	Pink/Black/A2
DOUT21	11	Orange/Red/A3
DOUT22	31	Orange/Black/A3
DOUT23	12	Gray/Red/A3
DOUT24	32	Gray/Black/A3
DOUT25	13	White/Red/A3
DOUT26	33	White/Black/A3
DOUT27	14	Yellow/Red/A3
DOUT28	34	Yellow/Black/A3
DOUT29	15	Pink/Red/A3
DOUT30	35	Pink/Black/A3
DOUT31	16	Orange/Red/A4
DOUT32	36	Orange/Black/A4
-	17	Gray/Red/A4
-	37	Gray/Black/A4
-	18	White/Red/A4
*P24V (P24G)	38	White/Black/A4
*P24V (P24G)	19	Yellow/Red/A4
*P24V (P24G)	39	Yellow/Black/A4
*P24V (P24G)	20	Pink/Red/A4
*P24V (P24G)	40	Pink/Black/A4
FG	Case	Shield



*Power lines change according to the OUTPUT type. ( ) are for type-P.